QCMP | 경북대학교 물리학과

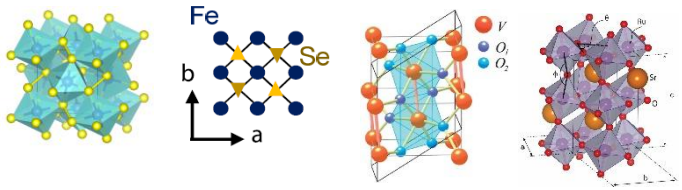
양자극한물성 연구실

"양자극한물성연구실은 **강상관계**에서 발현되는 여러 **양자 현상**을 공부합니다."

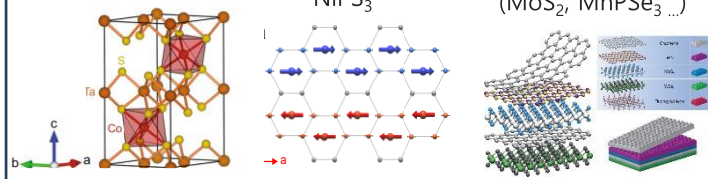
특이한 전자 또는 스핀구조를 가진 물질의 **양자 현상** 이해

01 물질군 분류

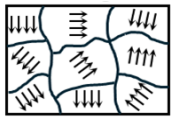
CoS_2 $FeSe/Te$ VO_2 $SrRuO_3$



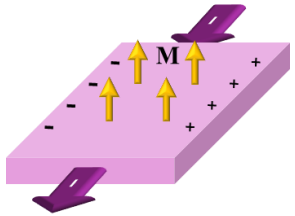
$Co_{1/3}STaS_2$ $FePS_3$ $MnPS_3$ $NiPS_3$ 2D vdW material (MoS_2, $MnPSe_3$ ...)



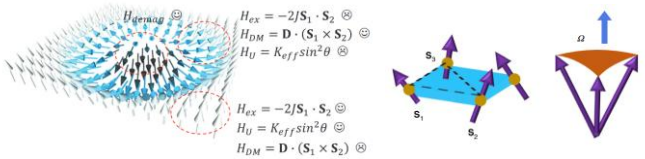
02 양자현상측정



자기 구조



양자 수송



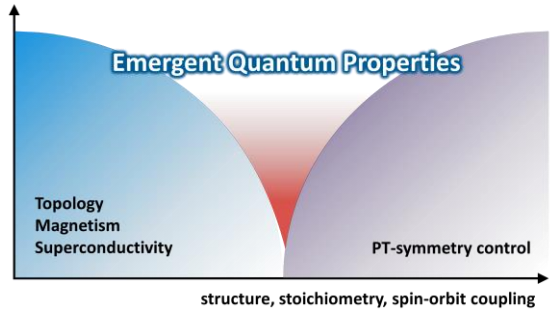
다양한 위상성질



박스 열고 슈뢰딩거 고양이 확인

대칭성 제어로 **새로운 양자상태** 발굴

03 대칭성 제어



물성의 **상호작용**과 **상전이** 간의 공존 원리를 규명하여 새로운 양자상태가 극대화 되는 물질을 **설계**하고 **발굴**

-양자현상(Quantum phenomena)
원자계의 **집단적 행동**으로 발현되는 자성, 초전도성, 위상 물성 등

지도교수

조연정 교수

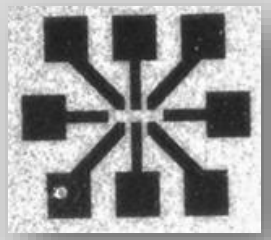


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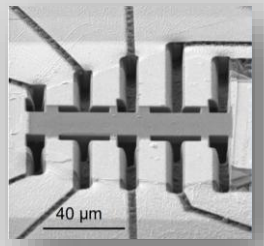
QCMP | 경북대학교 물리학과

양자극한물성 연구실

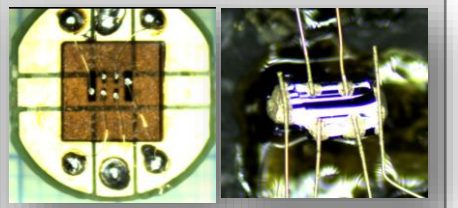
샘플 전처리



PhotoLithography



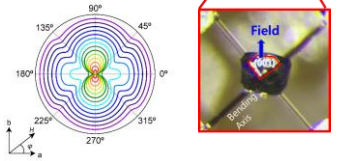
FIB Method

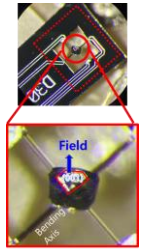


Wiring

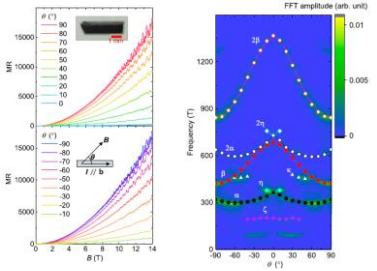
양자물성 측정

각도의존 토크
2차원 비등방성

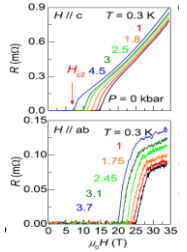




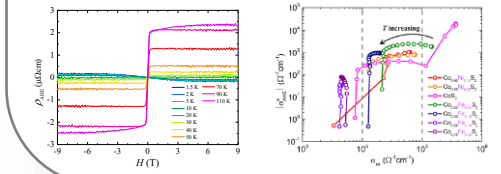
양자진동 전자구조, 베리phase



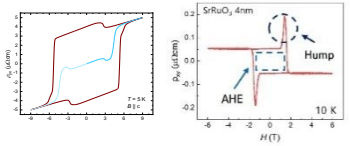
초전도체
임계특성



이상 홀전도도 베리곡률이해



Spin texture에 의한 위상 홀 효과



제어변수

기본물성

SEM XRD 계측장비



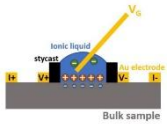
다중극한환경

고자기장
~41 T
(NHMFL)
9 T



극저온
1.6 K
(He_4)
300 mK
(He_3)

Ionic liquid gating

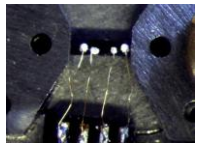


압력 및 스트레인

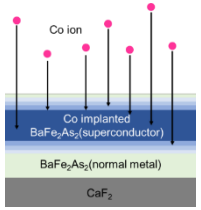


3GPa
(Indent)
1GPa
(CuBe)





Ion implantation



구인공고



학부인턴, 석사, 박사, 석박사통합 모집

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홈페이지 :
https://sites.google.com/view/labqcmp/